Self-Supervised Camera–Radar BEV Representation Learning with Semantic Distillation and Motion Preservation

DD2414 Engineering Project in Robotics, Perception and Learning
Project Specification

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1 Project Goal and Scope

Camera and automotive radar provide complementary information for autonomous driving perception. Cameras provide rich appearance and semantic information, while radar provides metric and dynamic information such as range, Doppler velocity and radar cross section. Existing camera–radar BEV systems such as Simple-BEV [1] can combine both modalities in a common bird's-eye-view representation, but the learned representation is normally optimized mainly for a downstream perception task.

In this project, we investigate whether a fused camera–radar BEV representation can be self-supervised to retain both **strong visual semantics** and **radar-specific motion information**.

We will use a BEV-style camera–radar architecture as the student. Radar Doppler and velocity information will already be encoded through a VoxelNet-style radar branch before fusion. A frozen DINOv2 [2] model will provide semantic supervision. In addition, a radar-derived motion task will encourage the fused representation to retain dynamic information that may otherwise become less prominent during fusion and semantic distillation.

Our design is partly inspired by CleverDistiller [3], which combines vision-foundation-model feature distillation with a separate sensor-specific self-supervised objective. We adapt this idea to a fused camera–radar BEV representation and use radar-derived motion information as the additional learning signal.

The main research question is:

> **Can DINOv2-based semantic distillation and radar-derived motion self-supervision jointly produce a fused BEV representation that preserves both semantic and dynamic information?**

We focus on **representation learning and evaluation**. It does not aim to design a new radar backbone, a new BEV geometry framework, or a complete autonomous-driving perception stack from scratch.

2 Technical Approach

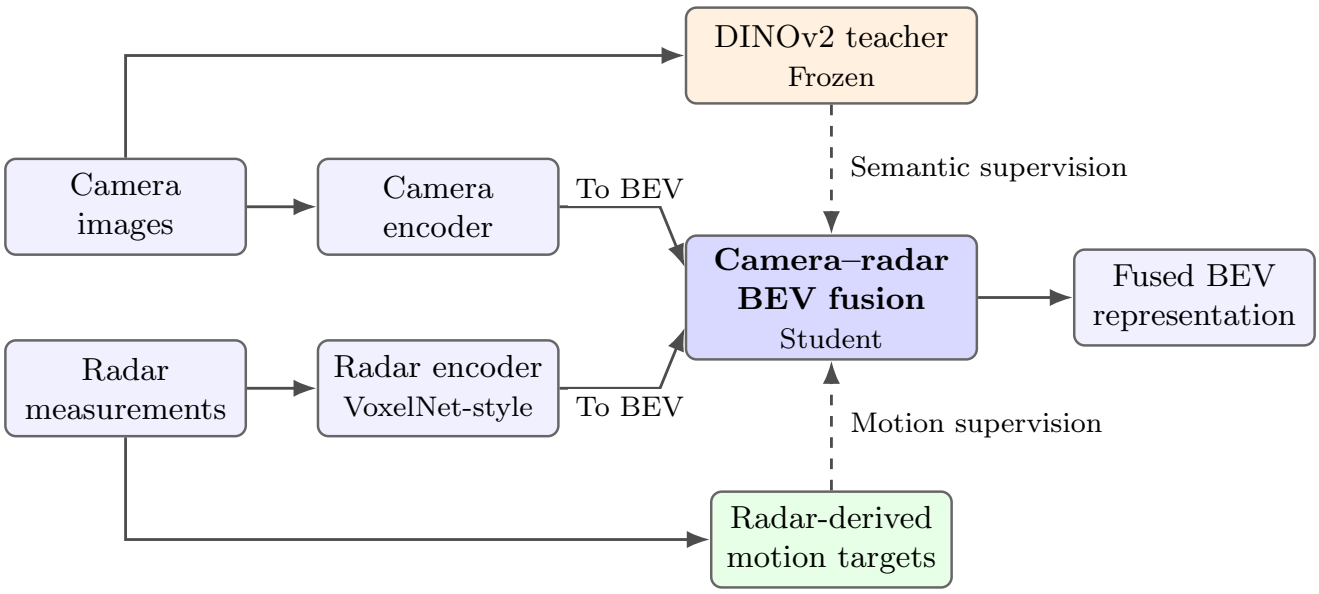


Figure 1: Overview of the proposed framework. Camera and radar features are mapped into BEV and fused into a shared student representation. A frozen DINOv2 teacher provides semantic supervision, while radar-derived targets provide motion supervision. Solid arrows indicate data flow; dashed arrows indicate training supervision. Correspondence operations and prediction heads are omitted for clarity.

We will reuse established components from Simple-BEV and related work where possible.

Camera branch. Camera images are encoded and projected into BEV using the geometric projection mechanism of Simple-BEV.

Radar branch. Inspired by the radar encoding approach in BEVCar [4], we will use a VoxelNet-style encoder to process radar measurements. Position, velocity and radar cross section (RCS) will be retained as input features, allowing motion information to enter the fused BEV representation through the radar branch.

Camera–radar fusion. Camera and radar BEV features are fused into a common BEV representation. This fused representation is the main student representation studied in the project.

Semantic distillation. The same camera images are processed by a frozen DINOv2 teacher. Using camera calibration and the geometric projection already available in the BEV pipeline, image-space DINO features are associated with corresponding BEV locations. A lightweight semantic projection head maps fused BEV features to the DINO feature space.

Planned contribution: soft-region correspondence. We plan to develop radar-supported soft-region correspondence for semantic distillation. Each projected radar return will be associated with a small region of DINOv2 features rather than a single image patch, aiming to reduce sensitivity to spatial uncertainty in radar–camera alignment.

Motion-preserving self-supervision. DoGFlow [5] uses radar Doppler measurements to derive motion pseudo-labels for self-supervised LiDAR scene flow learning. Inspired by this, we will investigate radar-derived supervision to preserve motion information in fused camera–radar BEV features. An auxiliary motion head will predict compensated radial velocity and/or moving-versus-static labels at radar-supported locations. Masked Doppler reconstruction will also be considered to reduce trivial copying of input velocity.

The overall training objective is conceptually

$$\mathcal{L}_{\text{pretrain}} = \mathcal{L}_{\text{semantic}} + \lambda \mathcal{L}_{\text{motion}},$$

where the first term encourages semantic knowledge transfer from DINOv2 and the second encourages radar-specific dynamic information to remain recoverable from the fused BEV representation.

The primary dataset will be nuScenes [6], which provides synchronized multi-camera images, automotive radar, ego-motion information and calibrated sensor poses.

3 Intended Learning Outcomes

In addition to the general DD2414 intended learning outcomes, the project defines the following project-specific outcomes.

After completing the project, we should be able to:

1. analyse a multimodal perception problem and select an appropriate technical approach under limited time and computational resources;
2. implement and debug a camera–radar BEV perception pipeline using existing research code and calibrated multimodal sensor data;
3. design and justify a self-supervised representation-learning objective for transferring complementary semantic and motion information;
4. design controlled experiments and ablations that distinguish the contribution of semantic distillation, radar motion information and their combination;
5. interpret experimental results, limitations and failure cases and use them to revise the project plan;
6. present and defend the technical solution, experimental design and conclusions both orally and in writing.

The first, fifth and sixth outcomes directly support the general course goals of planning and following up project work and of presenting and defending a technical solution in robotics, perception and learning.

4 Examination and Individual Assessment

The course consists of PRO1 and PRO2, each corresponding to 7.5 credits. Assessment will combine project progress, technical contribution, documentation and oral presentation.

PRO1 — first half of the project.

Expected evidence includes:

- an agreed project specification and updated work plan;
- a working nuScenes camera–radar data and BEV pipeline;
- a reproducible camera–radar baseline;
- initial DINOv2 integration and/or initial motion-learning prototype;
- a half-time presentation describing progress, technical decisions, problems and the plan for the second half.

PRO2 — second half of the project.

Expected evidence includes:

- the completed self-supervised training pipeline;
- systematic experiments and ablation studies;
- downstream and representation-level evaluation;
- final source code and experiment documentation;
- a written technical report;

- a final oral presentation and technical defense.

As a group project, each of us will have an identifiable technical contribution. Individual assessment can be based on code commits, experiment records, assigned technical work packages, report sections, presentation contributions and individual answers during technical discussions or presentations.

Both of us are expected to understand and be able to explain the overall system, even if implementation responsibilities are divided.

5 Work Plan

The project has two phases. The plan may be adjusted with the supervisor.

Stage	Planned work and expected output
PRO1: Baseline and feasibility	• Set up nuScenes and reproduce the BEV baseline. • Integrate the radar encoder with Doppler/velocity inputs. • Connect DINOv2 and establish initial BEV correspondence. • Present initial results and update the plan.
PRO2: Main study	• Implement semantic distillation and motion self-supervision. • Develop and evaluate soft-region correspondence. • Compare training variants and evaluate BEV representations. • Explore multi-sweep radar if time permits. • Complete the code, report and presentation.

The key controlled comparisons are expected to include:

- standard camera–radar BEV baseline;
- DINO semantic distillation only;
- radar motion self-supervision only;
- DINO semantic distillation + motion self-supervision;
- optionally, direct versus masked Doppler reconstruction.

A lightweight probe on frozen BEV features will be used where possible to test whether semantic and motion information are actually encoded in the learned representation rather than only recovered after full fine-tuning.

6 Expected Output and Main Risks

The expected technical output is a reproducible camera–radar BEV self-supervised pretraining pipeline and an experimental analysis of whether semantic distillation and radar motion supervision provide complementary benefits.

The expected course deliverables are:

- project specification and work plan;
- working research software;
- experiment logs and quantitative results;
- half-time presentation;
- final report;
- final presentation and demonstration where appropriate.

The main technical risks are DINO-to-BEV correspondence, radar sparsity, trivial Doppler reconstruction and competition between semantic and motion objectives. The initial implementation will therefore remain deliberately simple. Existing BEV geometry will be reused, and motion supervision will initially be applied only at radar-supported locations. Multi-sweep radar will remain an optional extension, subject to available time after the core method and evaluation are completed.

References

[1] A. W. Harley et al., "Simple-BEV: What Really Matters for Multi-Sensor BEV Perception?" arXiv:2206.07959, 2022.

[2] M. Oquab et al., "DINOv2: Learning Robust Visual Features without Supervision," TMLR, 2024.

[3] H. Govindarajan et al., "CleverDistiller: Simple and Spatially Consistent Cross-modal Distillation," arXiv:2503.09878, 2025.

[4] J. Schramm et al., "BEVCar: Camera-Radar Fusion for BEV Map and Object Segmentation," in *Proceedings of the IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)*, 2024, pp. 1435–1442.

[5] A. Khoche, Q. Zhang, Y. Cai, S. S. Mansouri, and P. Jensfelt, "DoGFlow: Self-Supervised LiDAR Scene Flow via Cross-Modal Doppler Guidance," *IEEE Robotics and Automation Letters*, vol. 11, no. 3, pp. 3836–3843, 2026.

[6] H. Caesar et al., "nuScenes: A Multimodal Dataset for Autonomous Driving," CVPR, 2020.